

Assignment 5: Structures and union in C

Set A . Write C programs for the following problems.

1. Create a structure employee (id, name, salary). Accept details of n employees and display it in summary format.

```
#include <stdio.h>
```

```
struct employee
{
    int id;
    char name[20];
    float salary;
};
struct employee e[100];
int main()
{
    int n,i;
    printf("Enter How many employees record you want : ") ;
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("Enter %d employee Record",i+1);
        printf("Enter Employee Id: ");
        scanf("%d",&e[i].id);
        printf("Enter Employee Name: ");
        gets(e[i].name);
        printf("Enter Employee Salary: ");
        scanf("%f",&e[i].salary);

    }
    printf("-----\n");
    for(i=0;i<n;i++)
    {
        printf("%d employee Record\n",i+1);
        printf("Employee Id is: %d\n",e[i].id);
        printf("Employee Name is: %s\n",e[i].name);
        printf("Employee Salary is: %f\n",e[i].salary);
    }
    return 0;
}
```

Output :

Enter How many employees record you want : 2

Enter 1 employee Record

Enter Employee Id: 10

Enter Employee Name: Ram

Enter Employee Salary: 20000

Enter 2 employee Record

Enter Employee Id: 40

Enter Employee Name: Alina

Enter Employee Salary: 25000

1 employee Record

Employee Id is: 10

Employee Name is: Ram

Employee Salary is: 20000.000000

2 employee Record

Employee Id is: 40

Employee Name is: Alina

Employee Salary is: 25000.000000

Set B . Write C programs for the following problems.

1. Create a structure item(item number, item name, rate, qty, total). Accept details of n items (calculate total as qty * rate). And display Bill in the following format.

Sr. No.	Item Name	Rate	Qty	Total
1	Pen	10	3	30.00
2	Eraser	5	1	05.00

Grand Total : 35.00

Prgramm :

```
#include <stdio.h>
#include <math.h>
struct item
{
    int id;
    char item_name[30];
    float rate;
    int qty;
    float total;
};
struct item items[100];
int main()
{
    int n,i;
    float grand_total=0;
    printf("Enter How many Items record you want : ") ;
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("Enter %d Item Record\n",i+1);
        printf("Enter Item Number: ");
        scanf("%d",&items[i].id);
        printf("Enter Item Name: ");
        gets(items[i].item_name);
        printf("Enter Rate: ");
        scanf("%f",&items[i].rate);
        printf("Enter Quantity: ");
        scanf("%d",&items[i].qty);
        items[i].total=items[i].rate*items[i].qty;
```

```

        grand_total=grand_total+items[i].total;
    }
    printf("-----\n");
    printf("Item No\tItem Name\tRate\tQuantity\tTotal\n");
    for(i=0;i<n;i++)
    {

printf("%d\t%s\t\t\t%0.2f\t%d\t\t%0.2f\n",items[i].id,items[i].item_name,items[i].ra
te,items[i].qty,items[i].total);
    }
    printf("-----\n");
    printf("Grand Total: \t%f",grand_total);
    return 0;
}

```

Output :

Enter How many Items record you want : 2

Enter 1 Item Record

Enter Item Number: 1

Enter Item Name: Pen

Enter Rate: 20

Enter Quantity: 6

Enter 2 Item Record

Enter Item Number: 2

Enter Item Name: Erase

Enter Rate: 5

Enter Quantity: 5

```

-----
Item No    Item Name    rate    Quantity    Total
1          Pen      20.00    6           120.00
2          Erase     5.00     5           25.00

```

```

-----
Grand Total:    145.000000

```

2. Create a structure car (car number, model name, colour, cost). Accept details of n cars and write a menu driven program to perform the following operations. Write separate functions for the different options.

i) Search by name

ii) Display all

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
struct car
{
    int car_num;
    char model_name[30],color[30];
    float cost;
};
struct car c[100];
void search_by_name(struct car c1[], int n);
void display_all(struct car c1[],int n);
int main()
{
    int n,i,choice;
    char cname[30];
    printf("Enter How many Cars record you want : ") ;
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("Enter Data of %d Car\n",i+1);
        printf("Enter Car Number: ");
        scanf("%d",&c[i].car_num);
        printf("Enter Car Model name: ");
        gets(c[i].model_name);
        printf("Enter Car Colour: ");
        gets(c[i].color);
        printf("Enter Car Cost: ");
        scanf("%f",&c[i].cost);
    }
    while(1)
    {
        printf("\n\n1.Search by name\n2.Display all\n3.Exit\n");
        printf("\nEnter Choice : ");
```

```

scanf("%d",&choice);
switch(choice)
{
    case 1: search_by_name(c,n);
            break;
    case 2: display_all(c,n);
            break;
    case 3: exit(0);
    default : printf("Invalid Choice");
}
}

void search_by_name(struct car c1[], int n)
{
    int flag=0,i;
    char cname[20];
    printf("Enter Car name whose details you required: ");
    gets(cname);

    for(i=0;i<n;i++)
    {
        if(strcmp(c1[i].model_name,cname)==0)
        {
            printf("\nDetails of Car %s is : \n",c1[i].model_name);
            printf("Car Number is %d\n",c1[i].car_num);
            printf("Car Color is %s\n",c1[i].color);
            printf("Car Cost is %0.2f\n",c1[i].cost);
            flag=0;
            break;
        }
        else
        {
            flag=1;
        }
    }
    if(flag==1)
        printf("Car Name is Not present in list\n");
}

```

```

void display_all(struct car c1[],int n)
{
    int i;
    for(i=0;i<n;i++)
    {
        printf("-----");
        printf("\nCar Name is %s\n",c1[i].model_name);
        printf("Car Number is %d\n",c1[i].car_num);
        printf("Car Color is %s\n",c1[i].color);
        printf("Car Cost is %0.2f\n",c1[i].cost);
    }
    printf("-----\n");
}

```

Output:

Enter How many Cars record you want : 2

Enter Data of 1 Car

Enter Car Number: 2345

Enter Car Model name: Alto

Enter Car Colour: red

Enter Car Cost: 400000

Enter Data of 2 Car

Enter Car Number: 6543

Enter Car Model name: Breeza

Enter Car Colour: White

Enter Car Cost: 900000

1.Search by name

2.Display all

3.Exit

Enter Choice : 1

Enter Car name whose details you required: Breeza

Details of Car Breeza is :

Car Number is 6543
Car Color is White
Car Cost is 900000.00

- 1.Search by name
- 2.Display all
- 3.Exit

Enter Choice : 2

Car Name is Alto
Car Number is 3243
Car Color is White
Car Cost is 400000.00

Car Name is Breeza
Car Number is 6463
Car Color is White
Car Cost is 900000.00

3. Create a structure movie (name, release year, duration). Accept details of n movies. Write a function to sort them according to release year. Display them in main() function. (Hint: Use dynamic memory allocation.)

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
struct movie
{
    int r_year;
    char movie_name[30];
    int duration;
};
struct movie *m;
void sort_movie(struct movie *m1,int n);
int main()
{
    int n,i,choice;

    printf("Enter How many Movie names you want to enter : ") ;
    scanf("%d",&n);
    m=(struct movie *)malloc(n*sizeof(struct movie));
    for(i=0;i<n;i++)
    {
        printf("\nEnter Movie Name : ");
        gets((m+i)->movie_name);
        printf("Enter Release year: ");
        scanf("%d",&(m+i)->r_year);
        printf("Enter Duration of Movie: ");
        scanf("%d",&(m+i)->duration);
    }
    sort_movie(m,n);
    printf("\n\nSorted Data According to Release Year is:\n");
    for(i=0;i<n;i++)
    {
        printf("\nMovie Name is : %s\n",(m+i)->movie_name);
        printf("Release year is : %d\n",(m+i)->r_year);
        printf("Duration of Movie is: %d\n",(m+i)->duration);
    }
}
```

```

void sort_movie(struct movie *m1, int n)
{
    int i,j;
    struct movie temp;

    for(i=0;i<n-1;i++)
    {
        for(j=0;j<n-i-1;j++)
        {
            if((m1+j)->r_year>(m1+j+1)->r_year)
            {
                temp=*(m1+j);
                *(m1+j)=*(m1+j+1);
                *(m1+j+1)=temp;
            }
        }
    }
}

```

Output:

Enter How many Movie names you want to enter : 2

Enter Movie Name : Agneepath

Enter Release year: 2012

Enter Duration of Movie: 2

Enter Movie Name : Dostana

Enter Release year: 2008

Enter Duration of Movie: 1

Sorted Data According to Release Year is:

Movie Name is : Dostana

Release year is : 2008

Duration of Movie is: 1

Movie Name is : Agneepath

Release year is : 2012

Duration of Movie is: 2